



Question Set

Question 1

Consider a Brownian risk factor used to drive an equity or rates model. Define the state variables (filtration, Brownian motion, adapted processes, quadratic variation, and stopping times) and derive the Ito dynamics, martingale condition, and exponential density process. State the measure, filtration, and tradability assumptions.

$$dX_t = \mu(t, X_t)dt + \sigma(t, X_t)dW_t, \quad d\langle W \rangle_t = dt$$

Question 2

For a Brownian risk factor used to drive an equity or rates model, explain which probability measure is natural for pricing. Write the density-process logic needed if the starting model is under the physical measure.

Question 3

Identify the discounted tradable or density-adjusted process in a Brownian risk factor used to drive an equity or rates model. Prove the martingale condition and name one modelling choice that would break it.

Question 4

Using a one-year Brownian path with monthly time steps and volatility 22 percent, construct a two-step numerical calculation. Show intermediate state values before discounting or aggregating.

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Question 5

For a Brownian risk factor used to drive an equity or rates model, compute and interpret drift removal and martingale exposure under a changed measure. Explain the hedge sign and the unit of each sensitivity.

Question 6

Suppose the desk observes drift and diffusion coefficients inferred from increments. Write the calibration objective, the weights you would use, and the parameter-identifiability checks.

Question 7

State the terminal and boundary conditions for a Brownian risk factor used to drive an equity or rates model. Explain how stopping times and path barriers defined by the available information prevents economically impossible prices.

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Question 8

Design a Monte Carlo or lattice estimator for a Brownian risk factor used to drive an equity or rates model. Specify Euler-Maruyama paths and quadratic-variation diagnostics, the estimator, the discounting rule, and the standard error.

Question 9

A junior quant implements a Brownian risk factor used to drive an equity or rates model and gets a stable but wrong answer. Explain how using future path information when defining a trading rule creates the error and design a diagnostic test.

Question 10

Write an NumPy algorithm for a Brownian risk factor used to drive an equity or rates model. Include inputs, outputs, diagnostics, and one benchmark case.

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Question 11

Starting from the payoff definition of a Brownian risk factor used to drive an equity or rates model, derive a replication, dynamic-programming, or conditional-expectation representation without skipping the economic argument.

Question 12

Create a stressed-market version of a one-year Brownian path with monthly time steps and volatility 22 percent. Recompute the value or risk metric and explain which term dominates the change.

Question 13

Use a second-order Taylor approximation for a Brownian risk factor used to drive an equity or rates model. Separate linear exposure, convexity exposure, cross terms if relevant, and residual model risk.

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Question 14

Show how a change of numeraire or change of measure would simplify one expectation in a Brownian risk factor used to drive an equity or rates model. Identify what becomes a martingale after the change.

Question 15

Give a proof-style answer showing when the wealth process for a Brownian risk factor used to drive an equity or rates model is self-financing. Explain why predictable trading strategies matter.

Question 16

Market quotes for drift and diffusion coefficients inferred from increments are noisy. Propose a robust calibration workflow with initial guesses, constraints, residual plots, and out-of-sample checks.

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Question 17

For a Brownian risk factor used to drive an equity or rates model, compare the mathematical boundary condition with the trader's intuition. Give one limiting case where the answer should be obvious.

Question 18

Design a convergence study for Euler-Maruyama paths and quadratic-variation diagnostics. Specify path counts, time-step refinement, confidence intervals, and the exact quantity plotted.

Question 19

Find one way that using future path information when defining a trading rule could pass a superficial unit test. Write a stronger test that would fail for the biased implementation.

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Question 20

Explain a Brownian risk factor used to drive an equity or rates model and code the core routine. Give the derivation outline, pseudocode, and three numerical sanity checks.